

B2's flaws, and where they show in its episodes

Last 150k training steps of each run (507 B2 episodes), decoded step by step from replay by tools/episode_cases.py. Everything is read from the agent's own observation and its own actions.

measured	B2	B1	honest	vanilla
tables placed per episode	2.22	2.00	1.51	0.75
tables placed with exactly 2 wood (all of it)	98%	54%	49%	63%
table-but-no-wood-pickaxe episodes left with 0 wood	100%	81%	64%	84%
stone-pickaxe chances per 100 episodes	19	13	5	2
pressed MAKE_STONE_PICKAXE when it was possible	6%	0%	0%	4%
chances lost to MAKE_WOOD_SWORD	36%	11%	10%	9%
pressed MAKE_WOOD_PICKAXE when it was possible	100%	13%	9%	15%
episodes that never drink	41%	37%	35%	45%
... drink reaches 0 at step (median)	205	200	205	189
lifespan: never drink / drink (median)	251 / 294	235 / 285	238 / 284	240 / 284
meals per episode	0.15	0.09	0.07	0.06
hostile adjacent in last 10 steps: attacks it	3%	2%	1%	2%
share of all steps that are DO	35%	23%	21%	18%
... aimed at grass	54%	58%	59%	64%

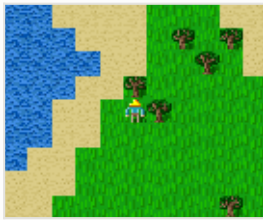
1. It spends its wood as fast as it gets it. B2 places a table 2.2 times per episode, almost always the instant it holds two logs -- which is all of them. 80% of the repeat tables are built with no table in view, so it builds a new one rather than walk back. Every episode that placed a table but never made the wood pickaxe was left with 0 wood afterwards, and never again stood at a table holding wood. At a table with wood it makes the pickaxe every time (the other runs: 9-15% of such steps) -- masking taught the key; the order of operations is not. B2 is the most extreme of the runs here; masking makes PLACE_TABLE pressable exactly when the second log arrives.
2. The stone-pickaxe chance lasts one step. B2 creates far more chances than any other run, but the median chance is 1 step long: in 36% the single log goes into the wood sword (also valid, also rewarded once), in 41% it walks off, and 17% succeed. Stone goes to furnaces (54%) and placed stone (44%), not tools.
3. The meters run it out. 41% of episodes never drink once; their drink hits zero around step 205 and they die around 251. It eats 0.15 times per episode. When drink hits zero, water is in view only 37% of the time -- it does not head for water it has seen before. With a hostile mob adjacent at the end it attacks it 3% of the time and NOOPs 20%.
4. A third of its steps are DO (35%), over half of them at grass (54%) -- a 10% sapling lottery; another 11% hit plants it placed. Long walks into a solid tile are rare (0.7 per 1000 steps).

In the episodes

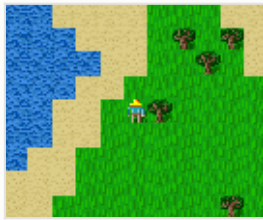
Frames are what the agent saw (its 9x11 view, dark tiles black), with the action it took next. Special actions in blue. From B2 replay.

Flaw 1: At the table with wood and stone; makes the wood sword instead

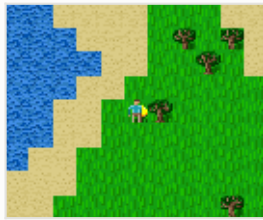
the sword spends its only wood; the pickaxe chance is gone



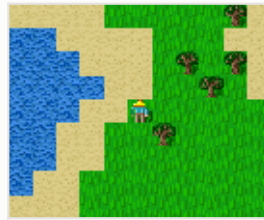
190: do
wood 2 stone 1
hp 9 food 2 drink 4



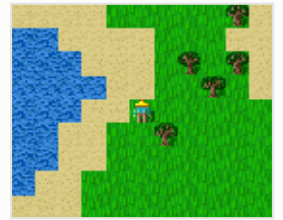
191: right
wood 3 stone 1
hp 9 food 2 drink 4



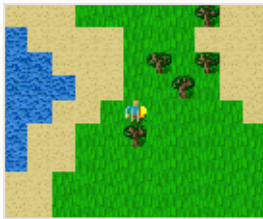
192: up
wood 3 stone 1
hp 9 food 2 drink 4



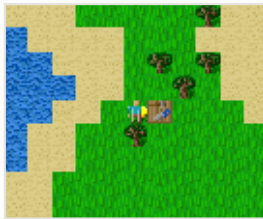
193: do
wood 3 stone 1
hp 9 food 2 drink 4



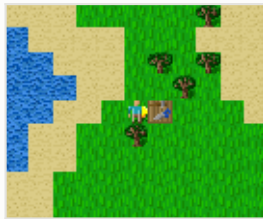
194: right
wood 3 stone 1
hp 9 food 2 drink 4



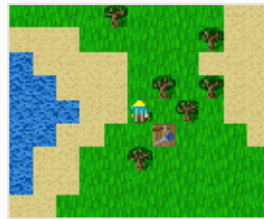
195: place table
wood 3 stone 1
hp 9 food 2 drink 4



196: make wood sword
wood 1 stone 1
hp 9 food 2 drink 4



197: up
wood 0 stone 1
hp 9 food 2 drink 4



198: right
wood 0 stone 1
hp 9 food 2 drink 4



199: do
wood 0 stone 1
hp 9 food 2 drink 4

Flaw 1: Holding both, a table in sight; walks off

every third step shown



126: up
wood 1 stone 1
hp 9 food 5 drink 4



129: left
wood 1 stone 1
hp 9 food 5 drink 4



132: up
wood 1 stone 1
hp 9 food 5 drink 4



135: left
wood 1 stone 2
hp 9 food 5 drink 4



138: do
wood 1 stone 2
hp 9 food 5 drink 3



141: right
wood 1 stone 2
hp 9 food 5 drink 3



144: left
wood 1 stone 2
hp 9 food 4 drink 3



147: noop
wood 1 stone 2
hp 9 food 4 drink 3



150: left
wood 1 stone 2
hp 9 food 4 drink 3



153: left
wood 1 stone 2
hp 9 food 4 drink 3

In the episodes

Frames are what the agent saw (its 9x11 view, dark tiles black), with the action it took next. Special actions in blue. From B2 replay.

Flaw 2: Places a second table beside the first

both logs go into another table; nothing left for the pickaxe



201: do
wood 1 stone 0
hp 9 food 2 drink 3



202: down
wood 2 stone 0
hp 9 food 2 drink 3



203: up
wood 2 stone 0
hp 9 food 2 drink 3



204: do
wood 2 stone 0
hp 9 food 2 drink 3



205: place plant
wood 2 stone 0
hp 9 food 2 drink 3



206: down
wood 2 stone 0
hp 9 food 2 drink 3



207: place table
wood 2 stone 0
hp 9 food 2 drink 3



208: sleep
wood 0 stone 0
hp 9 food 2 drink 3



209: right
wood 0 stone 0
hp 9 food 2 drink 3



210: right
wood 0 stone 0
hp 9 food 2 drink 3

Flaw 3: Drink at zero, water 1 tile away

it does not turn to the water



211: right
wood 1 stone 7
hp 9 food 2 drink 1



212: do
wood 1 stone 7
hp 9 food 2 drink 1



213: right
wood 1 stone 7
hp 9 food 2 drink 1



214: do
wood 1 stone 7
hp 9 food 2 drink 1



215: do
wood 1 stone 7
hp 9 food 2 drink 1



216: do
wood 1 stone 7
hp 9 food 2 drink 1



217: do
wood 1 stone 7
hp 9 food 2 drink 0



218: do
wood 1 stone 7
hp 9 food 2 drink 0



219: do
wood 1 stone 7
hp 9 food 2 drink 0



220: do
wood 1 stone 7
hp 9 food 2 drink 0

In the episodes

Frames are what the agent saw (its 9x11 view, dark tiles black), with the action it took next. Special actions in blue. From B2 replay.

Flaw 3: Killed beside a hostile mob

last steps of the episode



211: up
wood 0 stone 0
hp 4 food 2 drink 0



212: down
wood 0 stone 0
hp 4 food 2 drink 0



213: do
wood 0 stone 0
hp 4 food 2 drink 0



214: right
wood 0 stone 0
hp 2 food 2 drink 0



215: down
wood 0 stone 0
hp 2 food 2 drink 0



216: do
wood 0 stone 0
hp 2 food 2 drink 0



217: left
wood 0 stone 0
hp 2 food 2 drink 0



218: do
wood 0 stone 0
hp 2 food 2 drink 0



219: right
wood 0 stone 0
hp 2 food 2 drink 0



220: noop
wood 0 stone 0
hp 0 food 2 drink 0

Flaw 4: Walks into the same blocked tile

the move is repeated into a solid tile



152: right
wood 0 stone 0
hp 9 food 4 drink 3



153: right
wood 0 stone 0
hp 9 food 4 drink 3



154: right
wood 0 stone 0
hp 9 food 4 drink 3



155: right
wood 0 stone 0
hp 9 food 4 drink 3



156: right
wood 0 stone 0
hp 9 food 4 drink 3



157: right
wood 0 stone 0
hp 9 food 4 drink 3



158: right
wood 0 stone 0
hp 9 food 4 drink 3



159: right
wood 0 stone 0
hp 9 food 4 drink 3



160: right
wood 0 stone 0
hp 9 food 4 drink 3



161: left
wood 0 stone 0
hp 9 food 4 drink 3